Option 4. Scheduling project activities with limited resources

Decision owner and action

A project manager schedules activities that compete for crews and equipment, such as the tasks in a construction project or a plant shutdown. The tool chooses a start day for every activity. Deterministic starts are whole working days, and the checker rejects a fractional start.

Why this matters

The Critical Path Method (CPM) schedules each activity as early as its predecessors allow, but it assumes unlimited crews and equipment. On a real project, two activities that CPM starts on the same day may need the same crane or more electricians than are on site, so the schedule is impossible to execute and the delays spread through every successor. The project manager needs a schedule that respects both precedence and resource limits while finishing the project, and its milestones, as early as possible.

Files in the data package

File	Contents
`activities.csv`	One row per activity with duration in working days, an optional milestone due day, and a priority weight for milestone delay.
`precedence.csv`	One or more rows per predecessor relationship. A repeated row adds no requirement.
`resource_requirements.csv`	Units of each resource an activity occupies for its whole duration.
`resource_capacities.csv`	Units of each resource available on every day.

Required constraints

1. Every activity runs exactly once, without interruption, for its full duration.
2. An activity starts no earlier than the completion of every predecessor.
3. At every moment, the units of each resource used by running activities do not exceed its capacity. An activity that completes at time t releases its resources before an activity that starts at t uses them.

Required output

A `schedule` list with one entry per activity containing `activity_id`, `start`, and `completion`.

Measures

The primary measure is project completion, i.e., the latest activity completion. Secondary measures are weighted milestone delay (the sum, over activities with a milestone, of priority weight times days late, where days late is $\max(0, \text{completion} - \text{milestone due day})$), the number of late milestones, the gap in days between project completion and the CPM duration, and computation time. The CPM duration is a lower bound, i.e., a value no feasible schedule can beat, and not a feasible schedule, because the opening activities alone exceed the crew capacity.

Rule to beat

Serial scheduling by latest start. First compute CPM latest start times while ignoring resources. An activity's earliest start is the largest earliest completion of its predecessors (0 if none), and the CPM duration is the largest earliest completion. Working backward, an activity without successors has the CPM duration as its latest finish, any other activity's latest finish is the smallest latest start among its successors, and latest start is latest finish minus duration. Then repeatedly choose, among unscheduled activities whose predecessors are all scheduled, the one with the smallest CPM latest start (ties by activity identifier). Place it at the earliest time, no earlier than all

predecessor completions, at which every resource stays within capacity for its entire duration. If any activity requires more units of a resource than that resource's capacity, the rule stops and reports the instance infeasible.

Method directions and reading

A baseline may implement the rule to beat or use a different priority rule in the same serial procedure. A researched method may use a genetic algorithm or forward-backward improvement over activity lists, or an optimization model for smaller cases. Kolisch and Hartmann (1999, in *Project Scheduling: Recent Models, Algorithms and Applications*) classify these heuristics, and Hartmann and Kolisch (2000, *European Journal of Operational Research*) compare them. The CPM meeting comes after teams choose their options and less than two weeks before the prototype, and the rule to beat already needs CPM latest start times, so teams choosing this option should learn latest-start calculations from one of these sources first.

Uncertainty and replanning

Suitable uncertain inputs are activity durations and a temporary drop in crew capacity. The fixed-plan execution rule lists activities by planned start (ties by activity identifier) and gives each, in that order, the earliest time that is no earlier than the realized completion of every predecessor and the start of the activity listed before it, and at which every resource stays within realized capacity for the whole realized duration. Realized durations may be fractional, so starts and completions in scenarios are continuous times, and a capacity drop announced for day d applies to the interval $[d, d+1)$. Each scenario records when every capacity drop is announced, no later than its day, a replanning policy may use a drop only from its announcement onward, and the README states the announcement rule. A capacity drop does not interrupt activities that have already started, and when their combined requirement exceeds the reduced capacity, each affected resource and day is recorded as a hard-constraint violation. A replanning policy may reschedule activities that have not started whenever an activity finishes late or a capacity drop is announced, using the reduced capacity for the announced days. A useful stability measure is the number of activities whose start moves by more than a stated number of days.

Scope boundary

The required model has fixed capacities and one way to perform each activity. Buying extra resources, alternative execution modes, and detailed staffing are outside the required scope.

Checked example

Applying the rule to beat to the hand-checkable case gives a final completion time of 15 days. The decisions and every recomputed measure appear in `worked-example/` in the data ZIP. Reproducing this value exactly is a useful first test of a team's data loading and checker.

Where the rest of the project is defined

The project guide (`team-project-guide.pdf`) defines what every decision tool must do, the milestones, the grading rubric, the contents of this data package, and the exact rules the instructor's checker applies. This brief adds only what is specific to option 4.